Bootable Containers? Why and how

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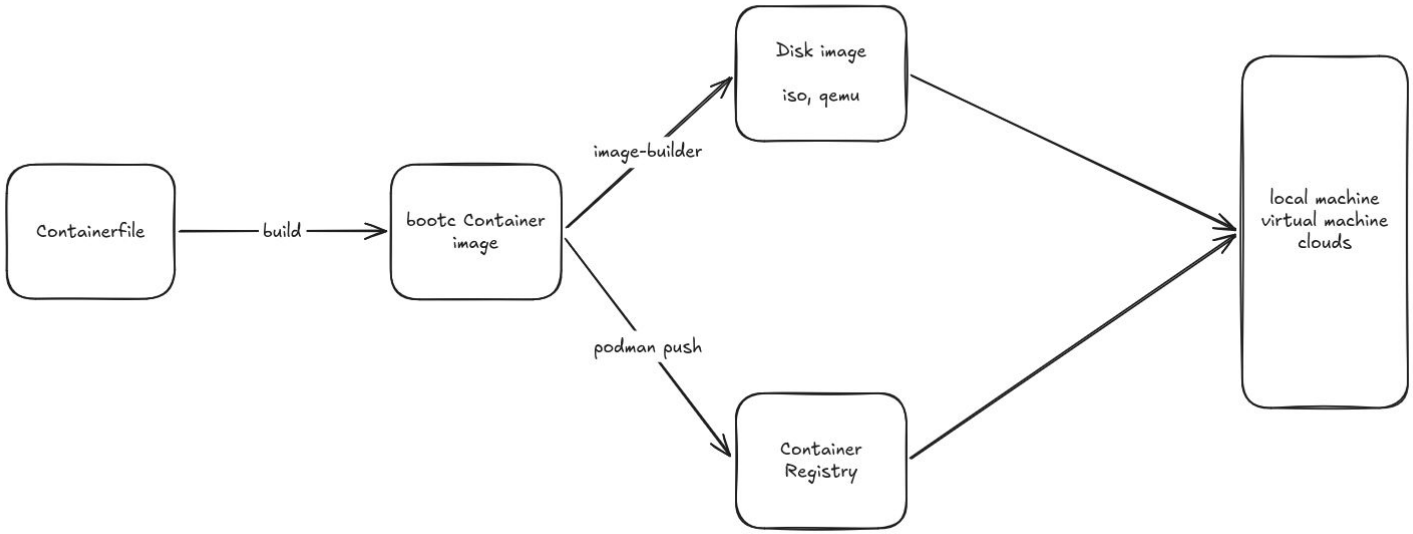


So, what is bootc?

- A container with the linux kernel
- What if you can use the same toolings that you use when dealing with container images to create operating systems?
  - Make use of container toolings to create an operating system
  - Make use of container registries to host os containers
  - Updates are just pulling from the registry
- Atomicity
- Immutable
- On boot, there is code in the initramfs to do a "chroot" equivalent into the target filesystem root



How it all works together



bootc

Building with bootc

– Containerfile

```
# The base image already has the linux
kernel.
FROM quay.io/fedora/fedora-bootc:44
RUN dnf install -y neovim firefox
RUN bootc container lint # Optional
```



Building with bootc

```
> podman build -t localhost/bootc-example
STEP 1/2: FROM quay.io/fedora/fedora-bootc:44
STEP 2/2: RUN dnf install -y neovim firefox
Updating and loading repositories:
 Fedora 44 - x86_64 - Updates Archive   100% |  11.4 MiB/s |  49.3 MiB |  00m04s
 Fedora 44 openh264 (From Cisco) - x86_ 100% | 823.0   B/s |   5.3 KiB |  00m07s
 Fedora 44 - x86_64 - Updates           100% |   1.4 MiB/s |  12.8 MiB |  00m09s
 Fedora 44 - x86_64                     100% |   3.1 MiB/s |  40.0 MiB |  00m13s
Repositories loaded.
.
.
.
Successfully tagged localhost/bootc-example:latest
275c05b8004fd5f999ac611d11c3001306ee5f4dc199649dafe87f42970348d1
```



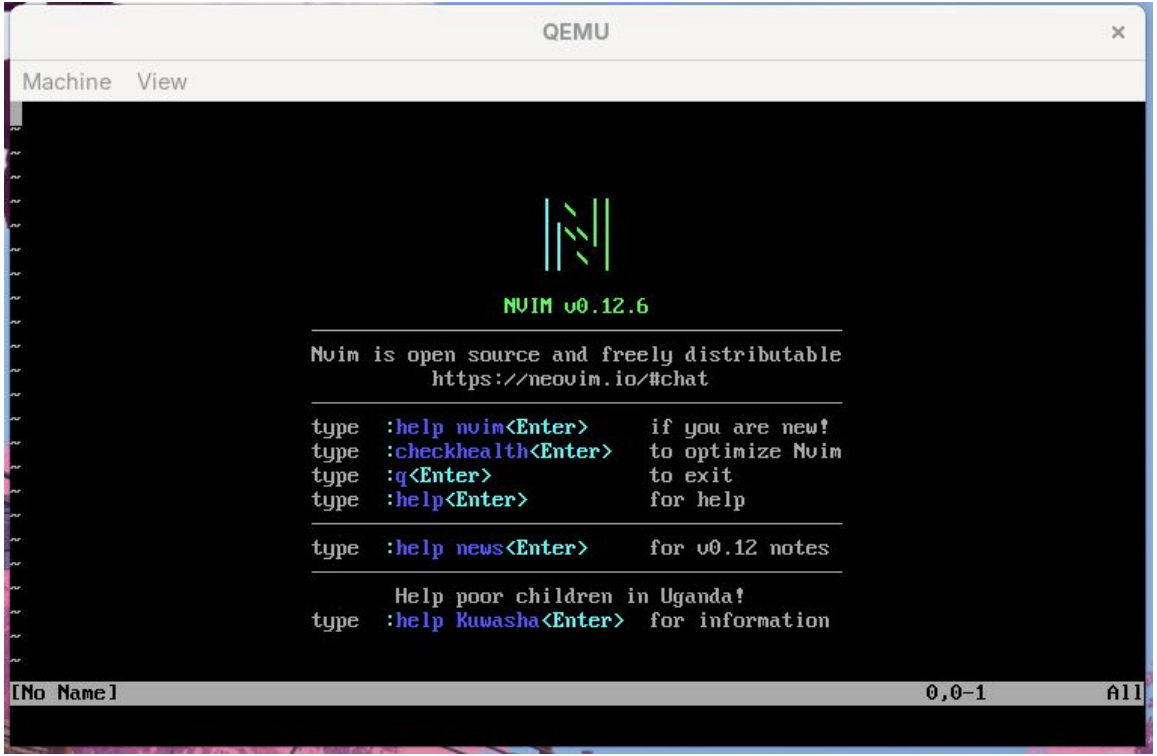
Build qemu artifact with image-builder

```
> podman run --rm --privileged --security-opt label=type:unconfined_t -v
/var/lib/containers/storage:/var/lib/containers/storage -v "$PWD/output:/output" -v "$PWD/config.toml:/config.toml:ro" \
quay.io/centos-bootc/bootc-image-builder:latest \
--type qcow2 --rootfs ext4 --config /config.toml quay.io/rchandar/bootc-example:latest

Generating manifest manifest-qcow2.json
Manifest generation step
Done generating manifest
Disk image building step
Building manifest-qcow2.json
starting -Pipeline source org.osbuild.containers-storage: cff93e445e0c252222a33808812872fccf7effe8fa97d43e507919fbebe323ad
Build
  root: <host>
.
.
.
  archive:      dc8f0659d4e34499062f0f83680f788017581e87db6a36eae6301f92901a1621
  gce:          0d2c038a9181bc036667847071bd63738987b782707a0e46107a9eec3a03aa82
Checkpoints
Build complete!
Results saved in .

> qemu-system-x86_64 -machine q35 -cpu host -m 4096 -smp 8 -enable-kvm -drive file=output/qcow2/disk.qcow2,format=qcow2   -device
virtio-vga-gl -display gtk,gl=on
```





neovim running in the bootc os

bootc

Bootc status inside the bootc os

```
bash-5.3# bootc status
● Booted image: quay.io/rchandar/bootc-example:v1.0
        Digest:
sha256:2804b20864f1948ab1770d5e977b5a17c172190f29f8264502702
caf115cf10d (amd64)
        Version: 44.20261102.0 (2026-11-03T16:40:07Z)
```



Uploading it to a container registry

```
> podman tag localhost/bootc-example quay.io/rchandar/bootc-example:latest
> podman image ls | grep bootc

localhost/bootc-example                    latest        275c05b8004f  46 minutes ago  3.88 GB
quay.io/rchandar/bootc-example             latest        275c05b8004f  46 minutes ago  3.88 GB

> podman push quay.io/rchandar/bootc-example:latest

Getting image source signatures
Copying blob 336d26816d85 [--------------------------------------] 8.0b / 50.3MiB | 1.6 MiB/s
Copying blob 336d26816d85 [--------------------------------------] 8.0b / 50.3MiB | 1.6 MiB/s
.
.
.
Writing manifest to image destination
```



Updating the containerfile

– Containerfile

```
FROM quay.io/fedora/fedora-bootc:44

# Add Apache2 server

RUN dnf install -y neovim firefox httpd

RUN systemctl enable httpd.service

RUN bootc container lint
```



Uploading the new image to the container registry

```
> podman tag localhost/bootc-example
quay.io/rchandar/bootc-example:latest

> podman push quay.io/rchandar/bootc-example:latest

Getting image source signatures
Copying blob 3a5eeccf8244 skipped: already exists
Copying blob 3a5eeccf8244 skipped: already exists
.
.
.

Writing manifest to image destination
```



upgrading to the new bootc container

```
bash-5.3# bootc status
·  Waiting for sysroot lock...
●  Booted image: quay.io/rchandar/bootc-example:latest
        Digest: sha256:2804b20864f1948ab1770d5e977b5a17c172190f29f8264502702caf115cf10d
(amd64)
        Version: 44.20261102.0 (2026-11-03T16:40:07Z)
bash-5.3# bootc upgrade
·  Waiting for sysroot lock…
layers already present: 65; layers needed: 3 (808.4 MB)
Fetched layers: 770.95 MiB in 7 minutes (1.72 MiB/s)
        Deploying: done (5 seconds)
        Queued for next boot: quay.io/rchandar/bootc-example:latest
        Version: 44.20261102.0
        Digest: sha256:44f06d32154e07523f056f693dedf4f29468eb05136fb206b0b3bddb63f97363
Total new layers: 68    Size: 1.8 GB
Removed layers:   66    Size: 3.9 GB
Added layers:     68    Size: 1.8 GB
```



after a reboot

```
bash-5.3# bootc status
⠁ Waiting for sysroot lock...
● Booted image: quay.io/rchandar/bootc-example:latest
        Digest:
sha256:44f06d32154e07523f056f693dedf4f29468eb05136fb206b0b3bddb63f97363
(amd64)
        Version: 44.20261102.0 (2026-11-04T10:25:53Z)
Rollback image: quay.io/rchandar/bootc-example:latest
        Digest:
sha256:2804b20864f1948ab1770d5e977b5a17c172190f29f8264502702caf115cf10d
(amd64)
        Version: 44.20261102.0 (2026-11-03T16:40:07Z)
        UpdateVersion: 44.20261102.0 (2026-11-04T10:25:53Z)
        UpdateDigest:
sha256:44f06d32154e07523f056f693dedf4f29468eb05136fb206b0b3bddb63f97363
```



Bootc in the wild

- Bazzite



- Fedora Atomic family



- Rhel Image Mode

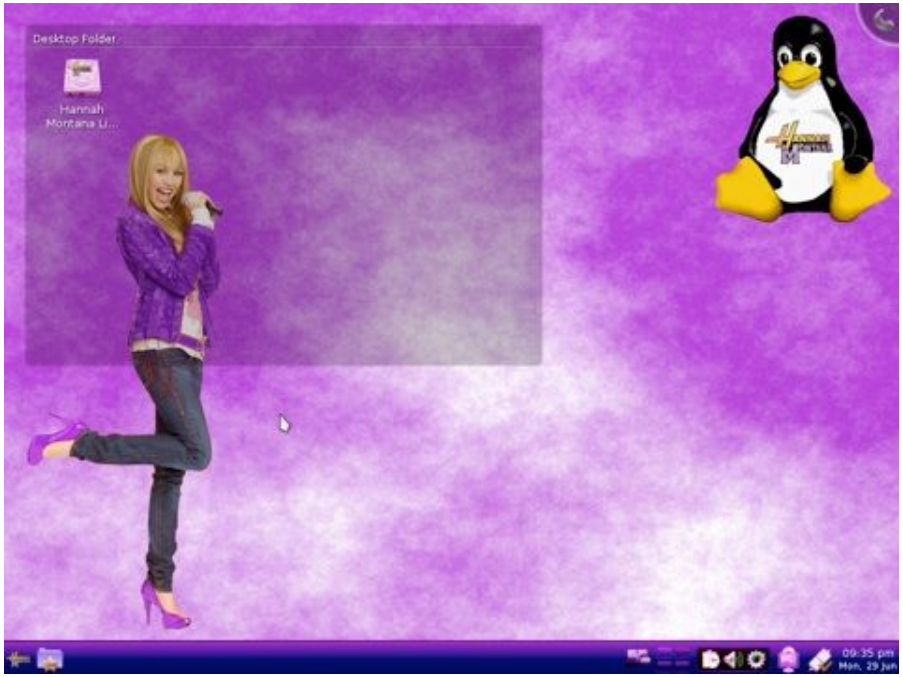


- Armada OS





Creating our own bootc os



Welcome to HannahMontana.sf.net



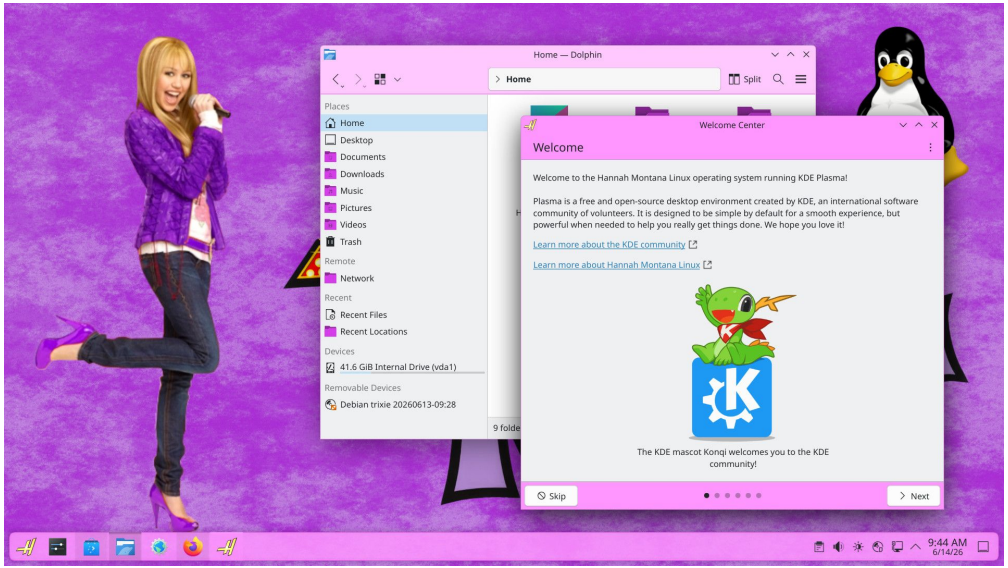
Hannah Montana Linux

Meet Hannah Montana Linux or HML for short. For more info, see the About page.

Hannah Montana Linux is a unix-like Linux Operating System based on Kubuntu.
The Package Manager is Debian apt.
The GUI is KDE 4.2 with Hannah Montana themes



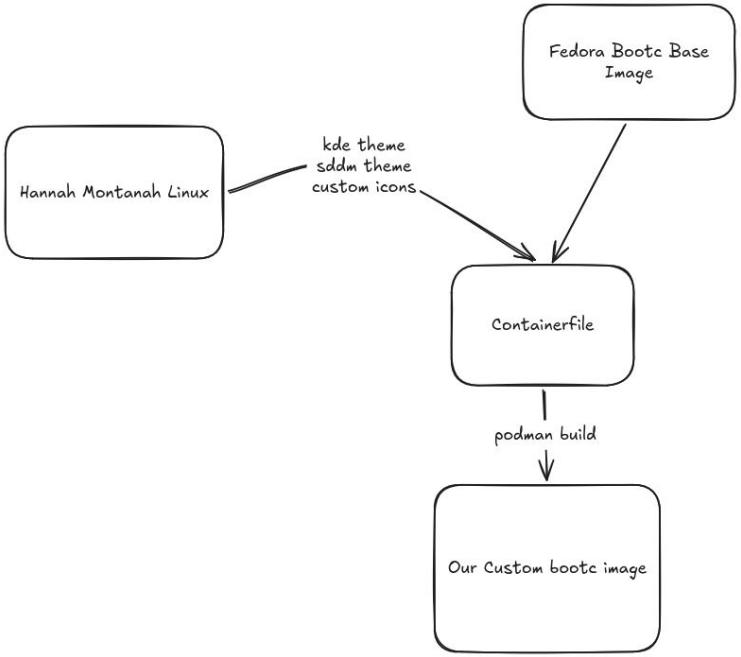
Hannah Montana Linux v26



- Based on the awesome debian live-build and calamares installer
- Uses modern plasma, some themes and a wallpaper. Easy to understand how to do

bootc

Game plan



1. Identify which files are the theme files
2. Create Containerfile with base as fedora bootc
3. Copy over the theme files in the containerfile
4. profit??



Containerfile

```
FROM quay.io/fedora/fedora-bootc:44

ARG VERSION=dev
ARG REVISION=unknown
LABEL org.opencontainers.image.title="Hannah Montana Linux" \
      org.opencontainers.image.description="An image-based KDE Plasma desktop powered by Fedora bootc" \
      org.opencontainers.image.licenses="GPL-3.0-or-later" \
      org.opencontainers.image.version="${VERSION}" \
      org.opencontainers.image.revision="${REVISION}" \
      containers.bootc="1"

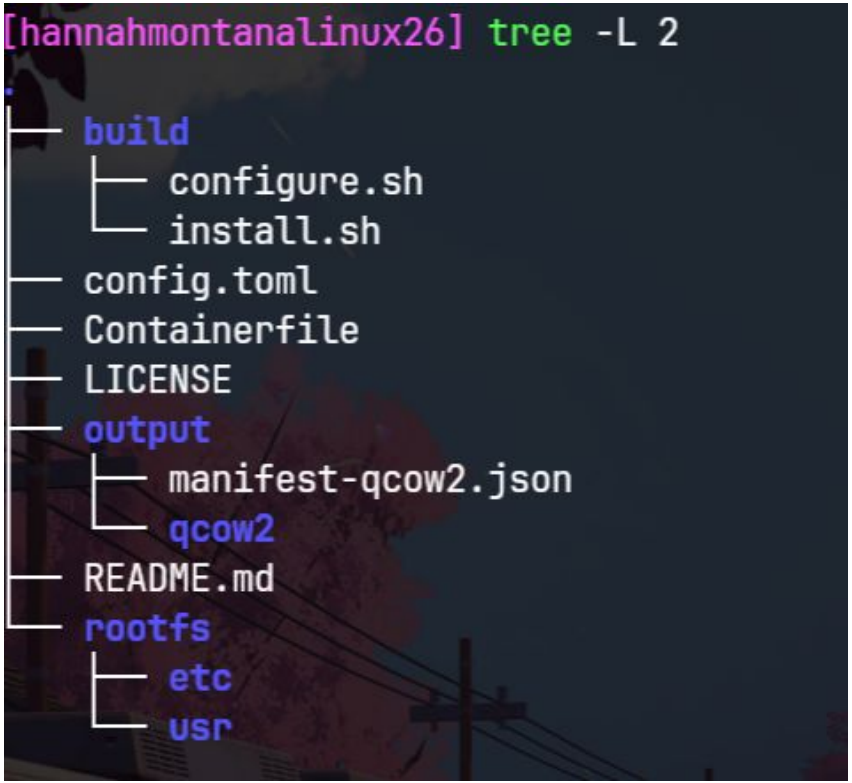
COPY build/ /run/build/
RUN bash /run/build/install.sh

COPY rootfs/ /

RUN bash /run/build/configure.sh
RUN rm -rf /run/build
RUN bootc container lint
```



File tree



bootc

install.sh

```
#!/usr/bin/bash

set -euo pipefail

dnf -y --setopt=install_weak_deps=False install plasma-desktop
plasma-workspace plasma-workspace-wayland kwin-wayland sddm sddm-breeze
polkit-kde systemsettings kde-gtk-config breeze-icon-theme
plasma-integration kscreen powerdevil xdg-desktop-portal-kde
google-noto-sans-fonts NetworkManager-wifi plasma-nm plasma-pa pipewire
pipewire-pulseaudio wireplumber pipewire-alsa bluez bluedevil
mesa-dri-drivers mesa-vulkan-drivers libglvnd-gles dolphin konsole kate
ark spectacle firefox flatpak fastfetch qt6-qt5compat qt6-qtdeclarative
qt6-qtsvg plymouth plymouth-plugin-script

dnf clean all
```



configure.sh

```
#!/usr/bin/bash
set -euo pipefail

# Use the installed Breeze QML with our artwork and theme
settings.
cp -a /usr/share/sddm/themes/breeze/*.qml \
    /usr/share/sddm/themes/hannah-montana-sddm/

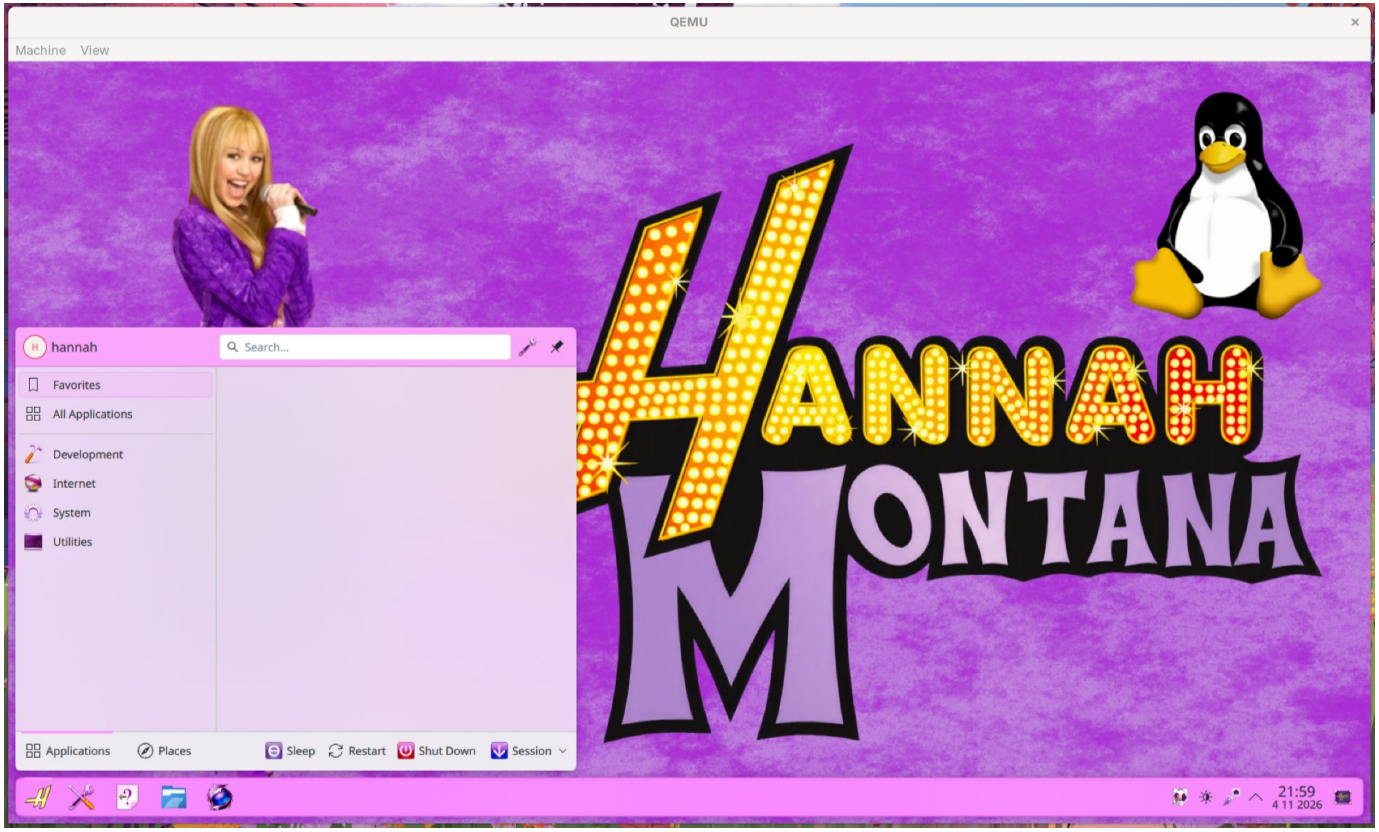
# Change display names, keeping Fedora's ID and version.
sed -i \
    -e 's/^NAME=.*/NAME="Hannah Montana Linux"/' \
    -e 's/^PRETTY_NAME=.*/PRETTY_NAME="Hannah Montana Linux
(Fedora bootc)"/' \
    /usr/lib/os-release

systemctl enable NetworkManager.service sddm.service
bluetooth.service

systemctl set-default graphical.target

plymouth-set-default-theme hannah-montana
```





our custom bootc image working as expected!

bootc

Some other cool features of bootc based Operating Systems

- Transactional updates, you can rollback from a bad update to a previous one
- Switch to another image, if you want to change from fedora atomic silverblue to fedora atomic kionite, just use bootc switch <image_url> :)



References

- https://www.youtube.com/watch?v=bf1xqjLeA9M
- https://gitlab.com/DecaCagle/hannahmontanalinux26
- https://www.youtube.com/watch?v=VKx5UZsX9jw
- https://github.com/Roshan-R/hannahmontanalinux-bootc



Questions?

github.com/Roshan-R

vazha.in/bootc-slides.pdf

